

In [2]:

```
import pandas as pd
import networkx as nx
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import math
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import mean_squared_error
from sklearn.metrics import mean_absolute_error
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
```

In [5]:

```
data = 'D:\\AI Course\\Bromley Work\\Copy of Copy of LBB - 12 Months INC.xlsx'
```

In [6]:

```
df = pd.read_excel(data)
print(df)
```

	IncidentID	ReportedSource	ReportedDate	\
0	INC000015101267	Email	2021-11-01 07:43:03	
1	INC000015100921	Phone	2021-11-01 08:11:14	
2	INC000015101286	Email	2021-11-01 08:43:25	
3	INC000015100937	Phone	2021-11-01 08:44:35	
4	INC000015101291	Phone	2021-11-01 08:49:49	
...	...	...	...	
15274	INC000016560283	Monitoring System	2022-11-22 18:58:28	
15275	INC000016560853	Monitoring System	2022-11-22 20:42:29	
15276	INC000016560869	Monitoring System	2022-11-22 20:42:29	
15277	INC000016559889	Monitoring System	2022-11-22 20:58:00	
15278	INC000016559532	Monitoring System	2022-11-22 21:27:27	
	IncidentClassification	\		
0	Normal			
1	Normal			
2	Normal			
3	Normal			
4	First Time Fix			
...	...			
15274	NaN			
15275	NaN			
15276	NaN			
15277	NaN			
15278	NaN			
	IncidentDescription	\		
0	Failed integration jobs			
1	Unable to connect to printer			
2	Issue with Software - Trapeze			
3	Unable to access adults liquid logic			
4	Unable to logon to citrix adp			
...	...			
15274	Device Reachability Problems on LBB2K8S248			
15275	Affected Application Class : NT_LOGICAL_DISKSR...			
15276	_DiscoveryStatus for NT_OS/NT_OS is in ALARM c...			
15277	ALERT: Bromley : LBB2K12S152 - Free Disk Space...			
15278	ALERT: Bromley : LBB5V557 - Database Backup Fa...			
	ResolutionNotes			
0	Failed jobs restarted			
1	spoken to user, after a reboot user can now pr...			
2	Re-licensed Trapeze Onstream on user laptop. S...			
3	If you're still unable to see the Adults Liqui...			
4	GromsD01User Unable to logon to citrix adpInco...			
...	...			
15274	NaN			
15275	NaN			
15276	NaN			
15277	NaN			
15278	NaN			

[15279 rows x 6 columns]

In [7]:

```
df
```

Out[7]:

	IncidentID	ReportedSource	ReportedDate	IncidentClassification	IncidentDescription	ResolutionNotes
0	INC000015101267	Email	2021-11-01 07:43:03	Normal	Failed integration jobs	Failed jobs restarted
1	INC000015100921	Phone	2021-11-01 08:11:14	Normal	Unable to connect to printer	spoken to user, after a reboot user can now pr...
2	INC000015101286	Email	2021-11-01 08:43:25	Normal	Issue with Software - Trapeze	Re-licensed Trapeze Onstream on user laptop. S...
3	INC000015100937	Phone	2021-11-01 08:44:35	Normal	Unable to access adults liquid logic	If you're still unable to see the Adults Liqui...
4	INC000015101291	Phone	2021-11-01 08:49:49	First Time Fix	Unable to logon to citrix adp	GromsD01User Unable to logon to citrix adpInco...
...	...	...	...	...	...	...
15274	INC000016560283	Monitoring System	2022-11-22 18:58:28	NaN	Device Reachability Problems on LBB2K8S248	NaN
15275	INC000016560853	Monitoring System	2022-11-22 20:42:29	NaN	Affected Application Class : NT_LOGICAL_DISKSR...	NaN
15276	INC000016560869	Monitoring System	2022-11-22 20:42:29	NaN	_DiscoveryStatus for NT_OS/NT_OS is in ALARM c...	NaN
15277	INC000016559889	Monitoring System	2022-11-22 20:58:00	NaN	ALERT: Bromley : LBB2K12S152 - Free Disk Space...	NaN
15278	INC000016559532	Monitoring System	2022-11-22 21:27:27	NaN	ALERT: Bromley : LBBSVS57 - Database Backup Fa...	NaN

15279 rows × 6 columns

In [8]:

```
df.isnull().sum()
```

Out[8]:

```
IncidentID          0
ReportedSource      0
ReportedDate        0
IncidentClassification  1886
IncidentDescription    1
ResolutionNotes     255
dtype: int64
```

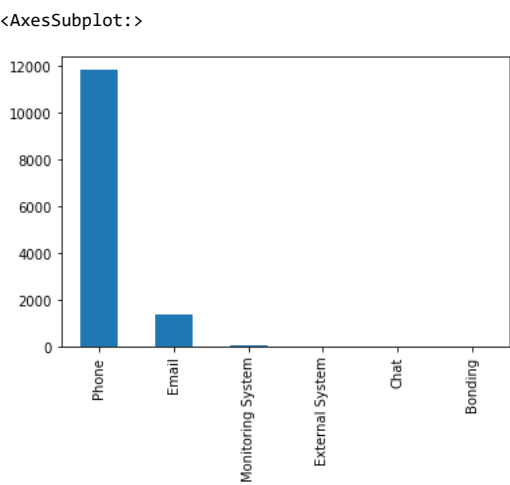
In [10]:

```
df.dropna(inplace=True)
```

In [13]:

```
c = df['ReportedSource'].value_counts()
c.plot(kind='bar')
```

Out[13]:



In [14]:

```
df.dtypes
```

Out[14]:

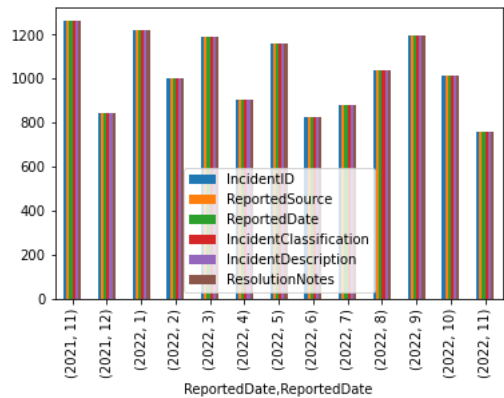
```
IncidentID          object
ReportedSource      object
ReportedDate        datetime64[ns]
IncidentClassification  object
IncidentDescription  object
ResolutionNotes      object
dtype: object
```

In [37]:

```
plt.figure(figsize=(100,50))
df.groupby([df["ReportedDate"].dt.year,df["ReportedDate"].dt.month]).count().plot(kind="bar")
```

Out[37]:

<AxesSubplot: xlabel= 'ReportedDate,ReportedDate' >
<Figure size 7200x3600 with 0 Axes>

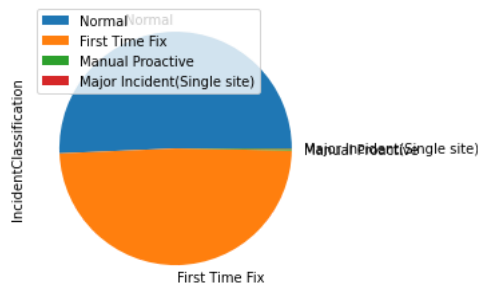


In [34]:

```
c = df['IncidentClassification'].value_counts()
c.plot(kind='pie')
plt.legend(loc='upper left')
```

Out[34]:

<matplotlib.legend.Legend at 0x1accb5d1b20>



In []: